19. Electrical Safety in the Medical Environment

한경대학교 생물정보통신전문대학원

김수찬

19. Electrical Safety

학습 목표

- 의료장비에서 전기적 안전에 대한 정의
- 의료 기관에서의 전기적 안전의 범위
- 전기적 안정성과 관련된 주요 기관
- 전기적 위험성을 줄일 수 있는 방법
- 인체의 전기에 대한 생리적 반응
- 악영향을 미치는 60Hz에서의 macroshock과 microshock
- Leakage current의 정의
- 전기적 위험성을 줄이기 위한 power isolation transformer system과 line isolation monitor에 관하여.
- Ground fault interrupter
- 전기 안전성 시험 장비

전기적 안전성의 정의

- Electric shock
  - Macroshoch –

  - Microshoch –

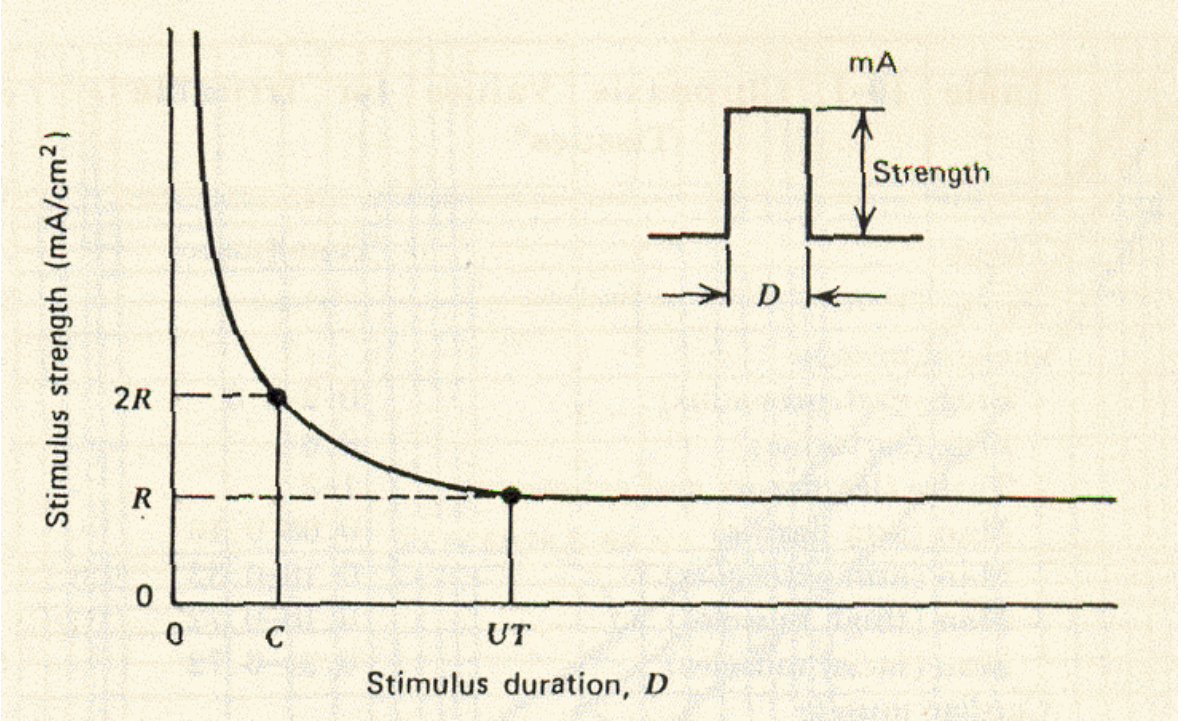
ㅏ.

- Explosion : 전기적 접촉에 의한 스파크로 인한 인화성 가스의 점화
- Fire : 부적절한 결선이나 과부하
- 장비나 건물의 손상

전기적 안전성에 대해 언급한 주요 기관들

- 주법
  - OSHA (Occupational Safety and Health Administration)
  - HEW (Health, Education and Welfare)
  - PHS (Public Health Service)
  - FDA (Food and Drug Administraiton)
- 자율 단체
  - NFPA (National Fire Protection Association)
  - NEC (National Electrical Code)
  - ANSI (American National Standards Institute)
  - UL (Underwriters Laboratories)
  - AAMI (Association for the Advancement of Medical Instrumentation)
  - JCAHO (Joint Commission on Accreditation of Healthcare Organizations)

전기 자극 강도와 자극 시간



C: Chronaxie –

R: Rheobase –

UT: Utilization time

인체에 전기를 가했을 때 생리학적 영향

- 조직에 상처
- 타의적인 근육 수축 혹은 의식 상실
- 심장 쇄동
- Macroshock과 microshock

인체에 전기를 가했을 때 생리학적 영향

Table. Effects of 60Hz electric shock (current) through the body of an average human

Current (1 second contact)	Physiological Effect
1 mA	
5 mA	
10-20 mA	
50 mA	Pain. Possible fainting, exhaustion, mechanical injury; heart and respiratory functions continue.
100-300 mA	Ventricular fibrillation will start, but respiratroy center remains intact. fatal if continued.
6A	

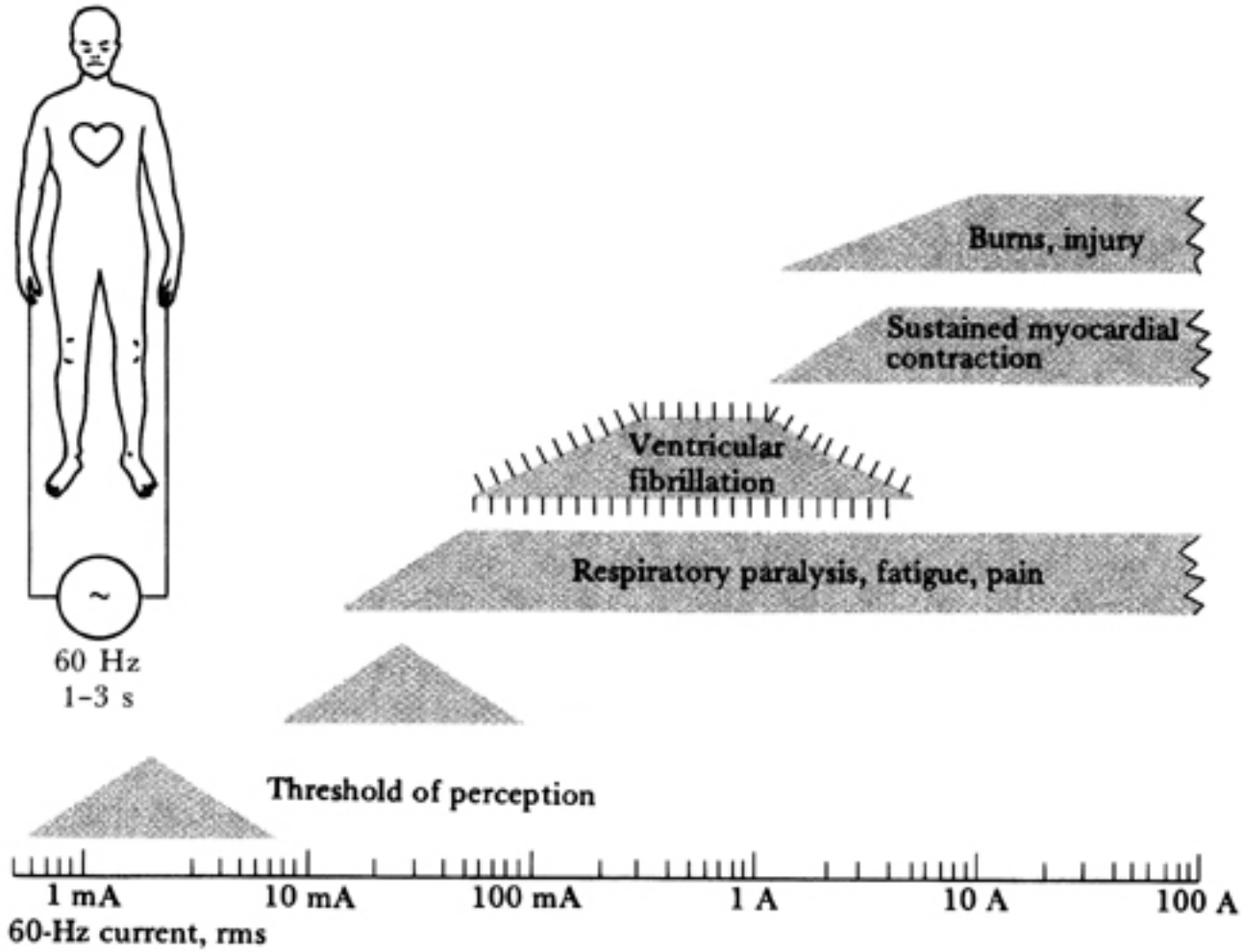
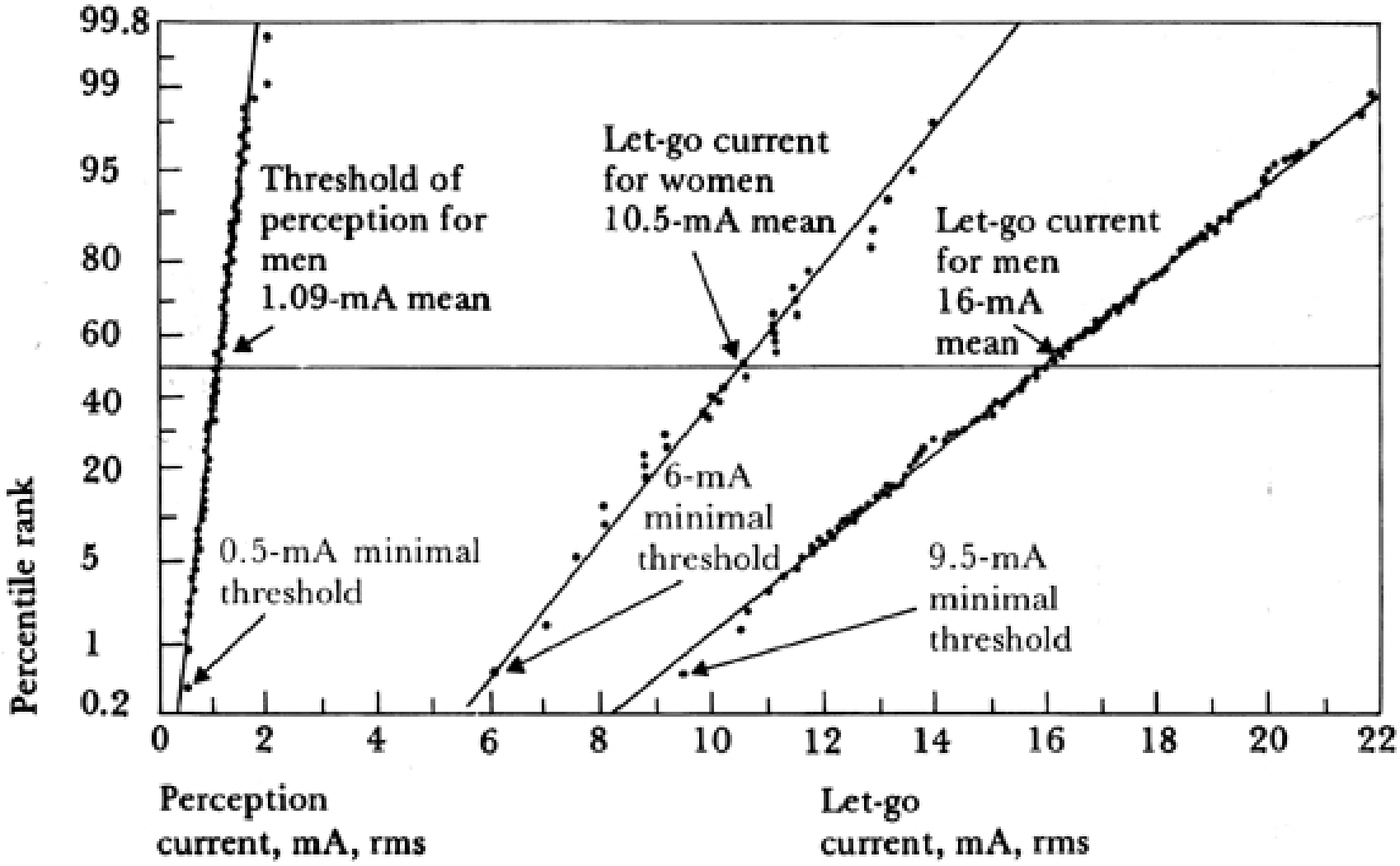


Figure. Threshold or estimated mean values are given for each effect in a 70 kg human for a 1 to 3 s exposure to 60 Hz current applied via copper wires grasped by the hands.

인체에 전기를 가했을 때 생리학적 영향



Let-go current versus frequency

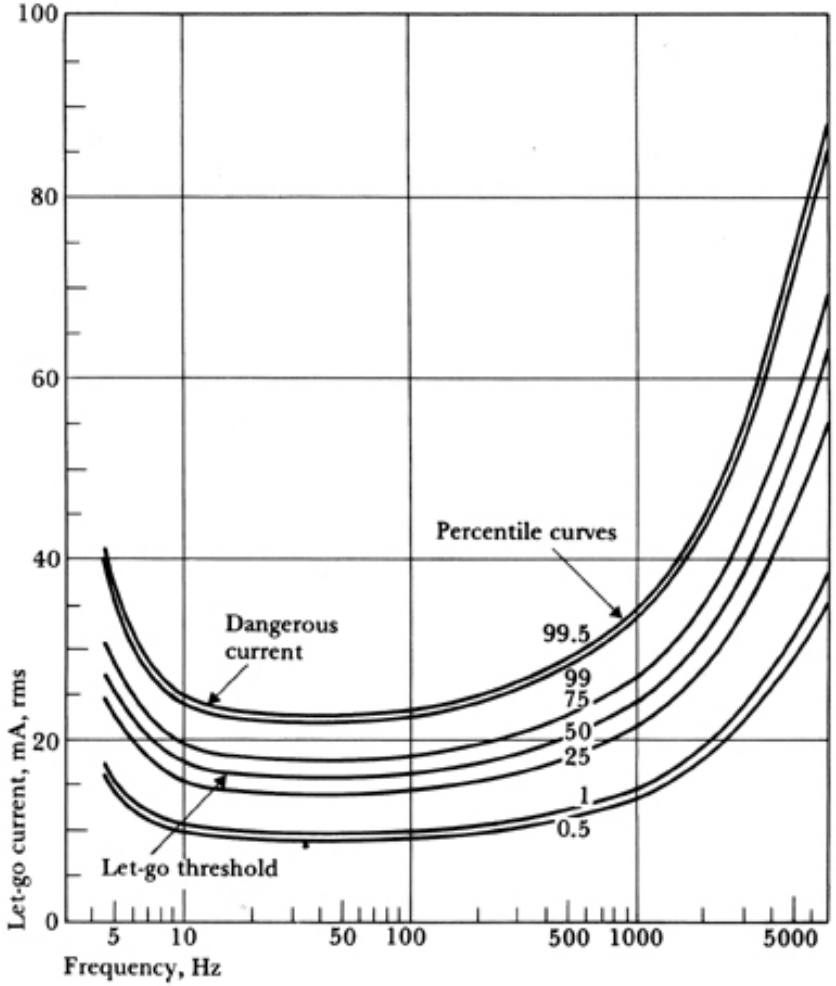


Figure. Percentile values indicate variability of let-go current among individuals. Let-go currents for women are about two-thirds the values for men.

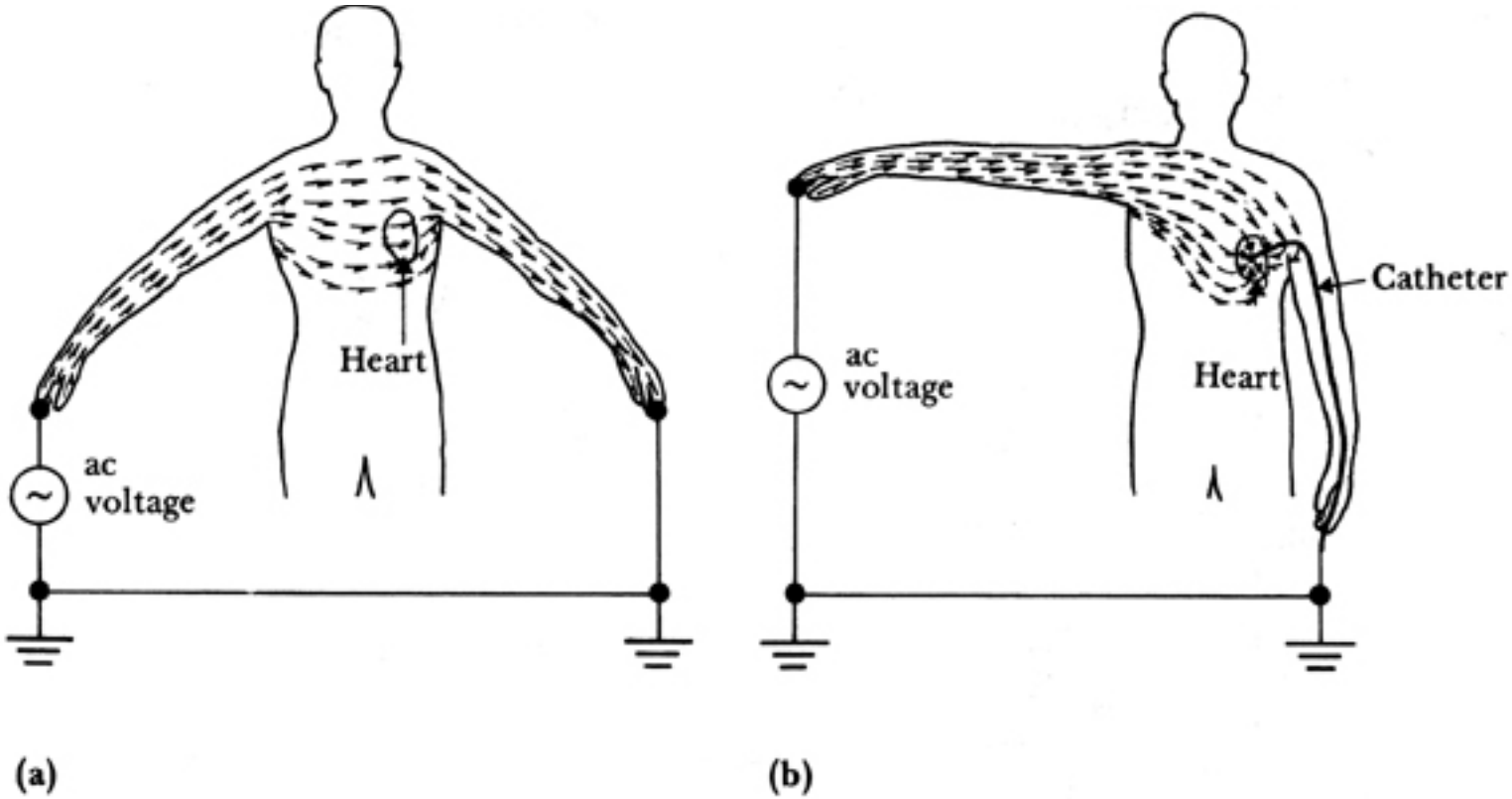


Figure. Effect of entry points on current distribution (a) Macroshock, externally applied current spreads throughout the body. (b) Microshock, all the current applied through an intracardiac catheter flows through the heart.

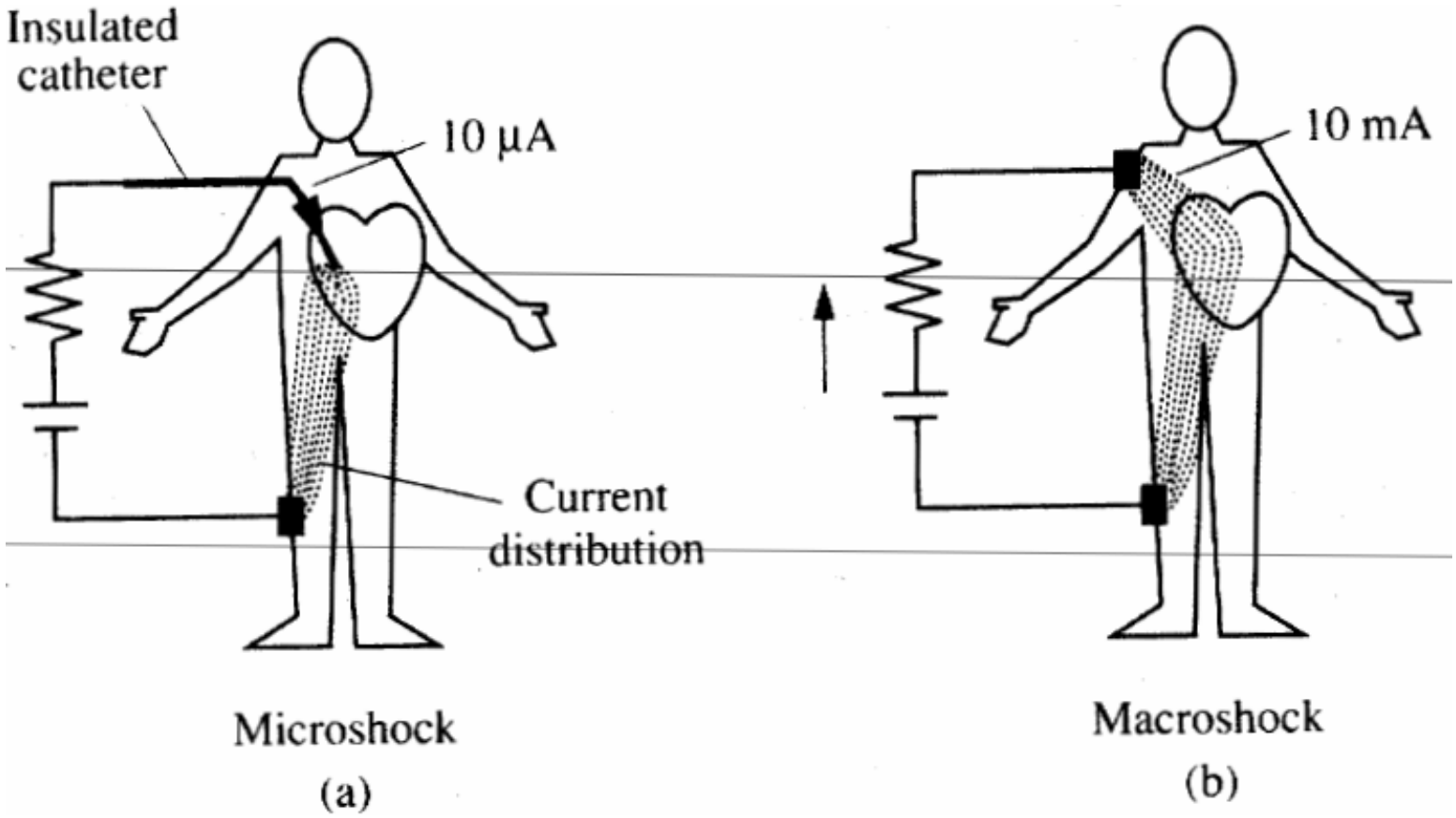


Figure. The current density accompanying microshock versus that accompanying macroshock

Macroshock과 microshock

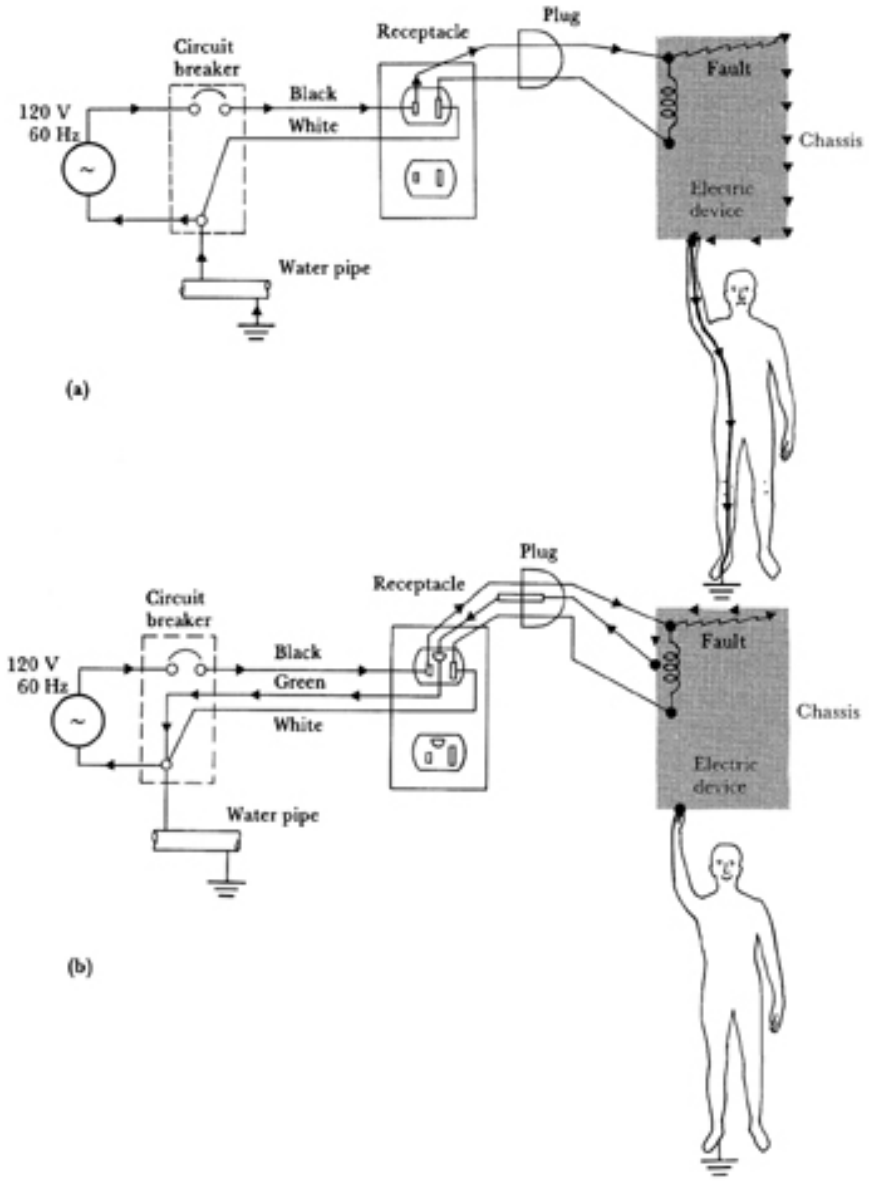


Figure. Macroshock due to a ground fault from hot line to equipment cases for (a) ungrounded cases and (b) grounded chassis.

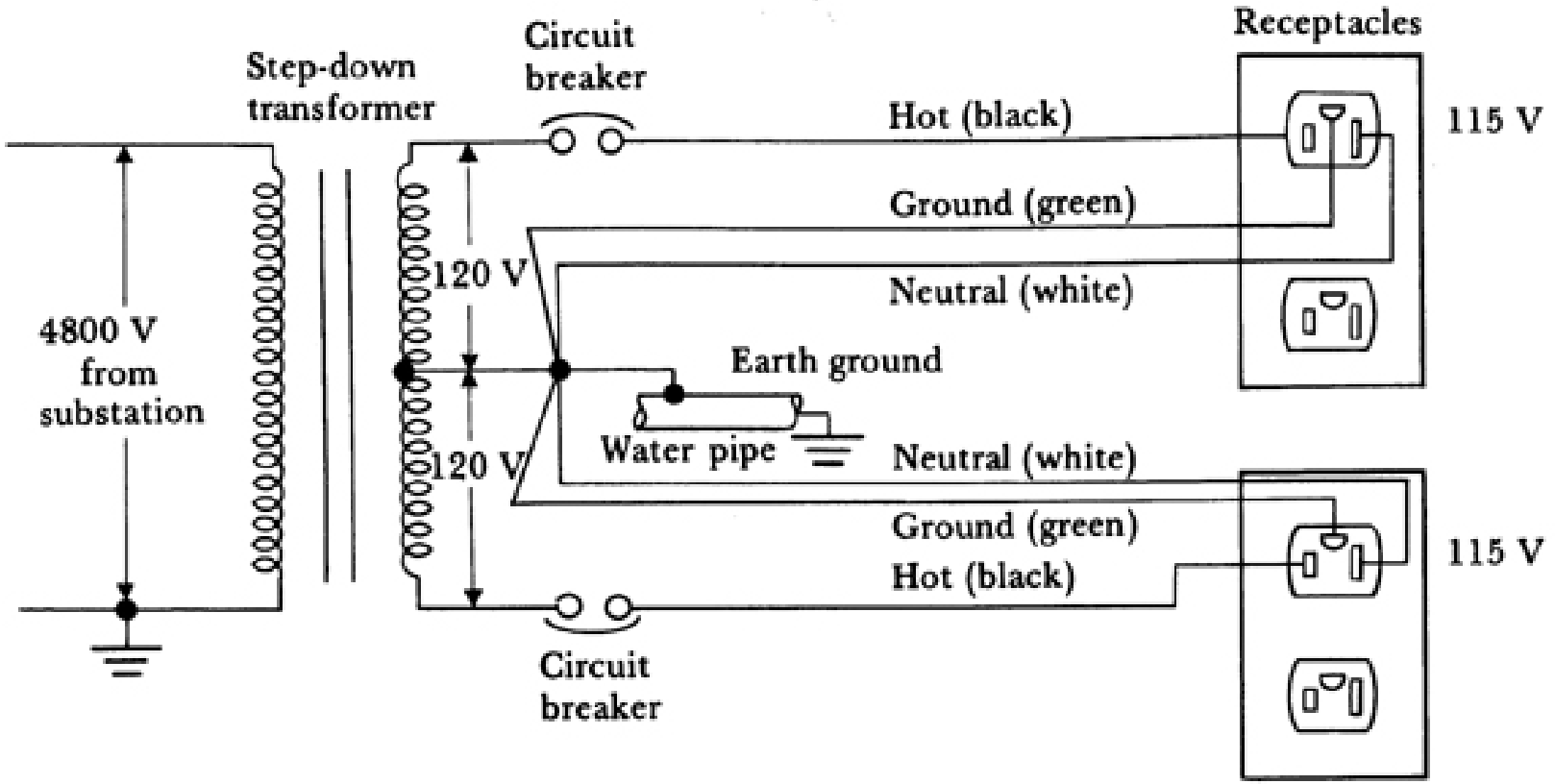


Figure. Simplified electric-power distribution for 115 V circuits. Power frequency is 60 Hz.

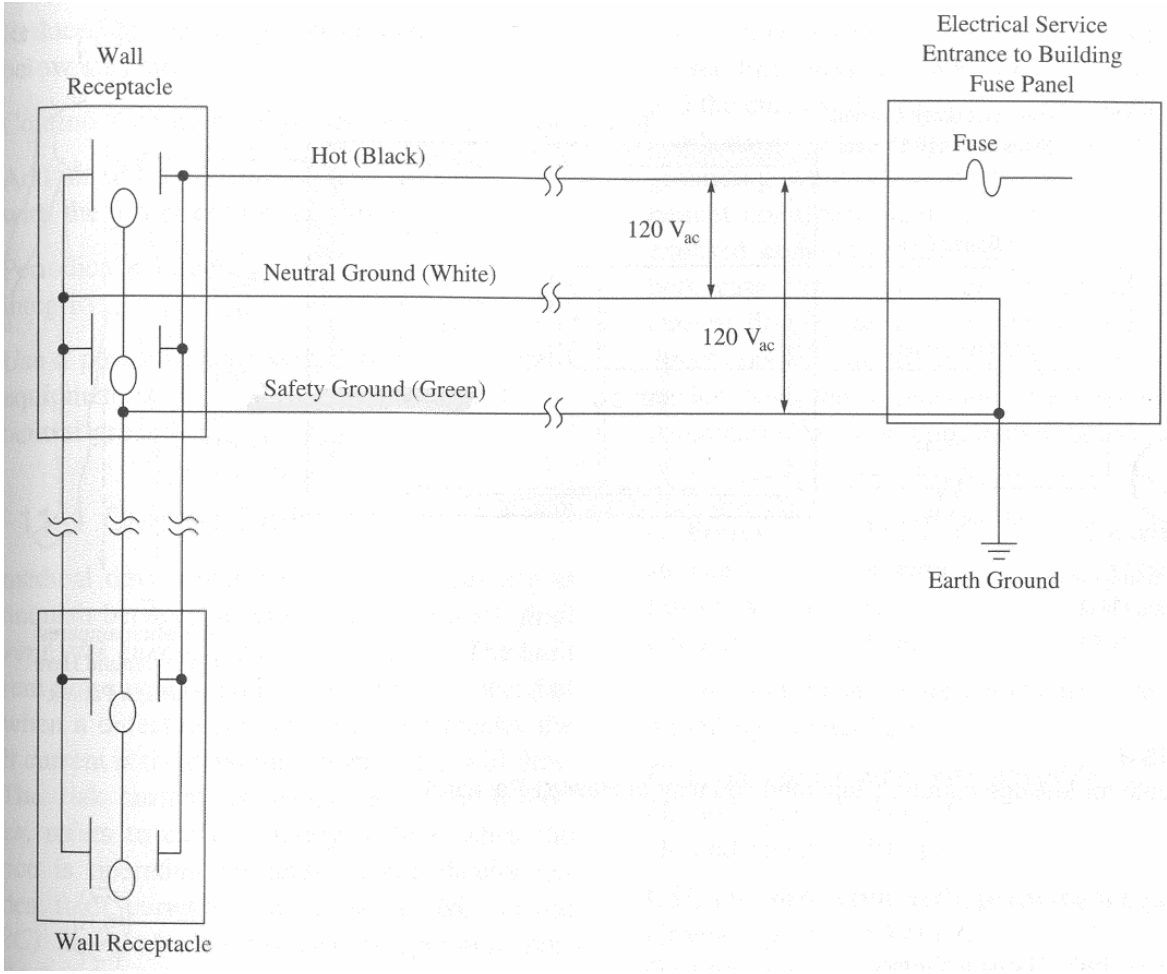


그림 15-7. 퓨즈 패널에 연결된 콘센트

누설전류 (Leakage current)

-

-

누설전류 (Leakage current)

- Capacitive leakage current
  -
    때문에 발생
- Resistive leakage current
  - 해서 발생
  - 용량성 누설 전류에 비해 무시할 만큼 작다.

누설전류 (Leakage current)

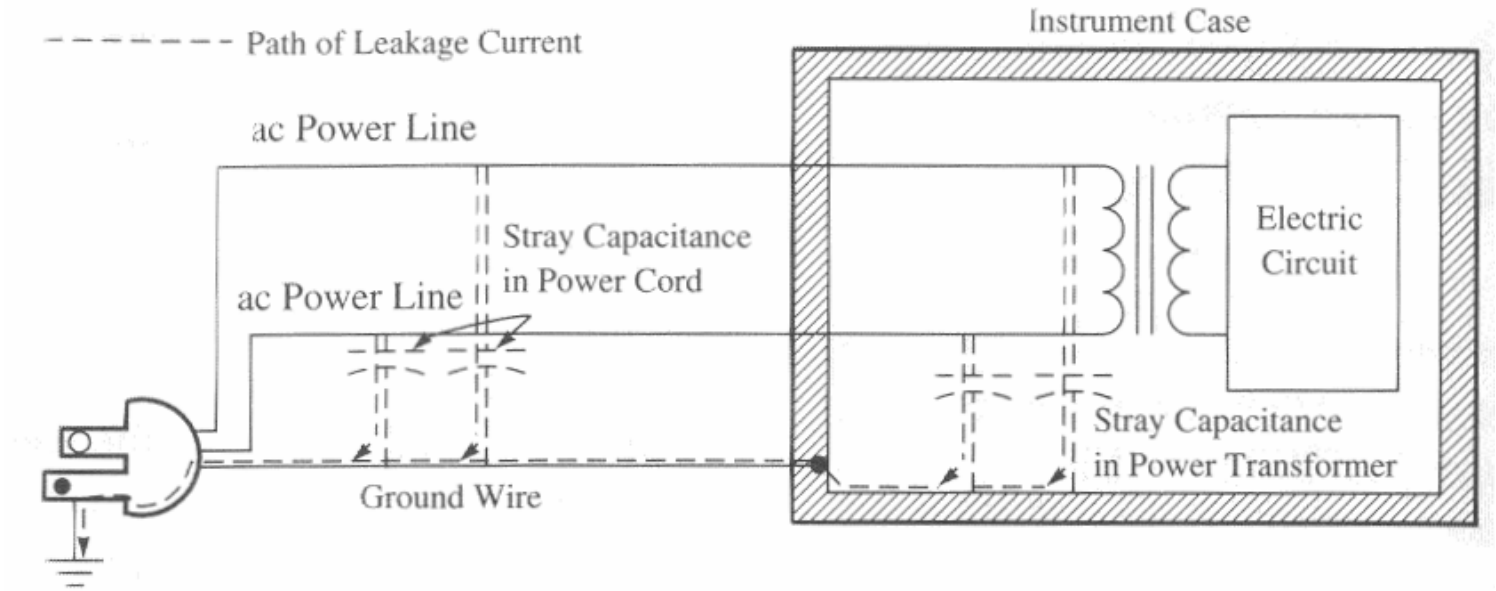


그림 15-5. 표유커패시터의 누설 전류의 원인

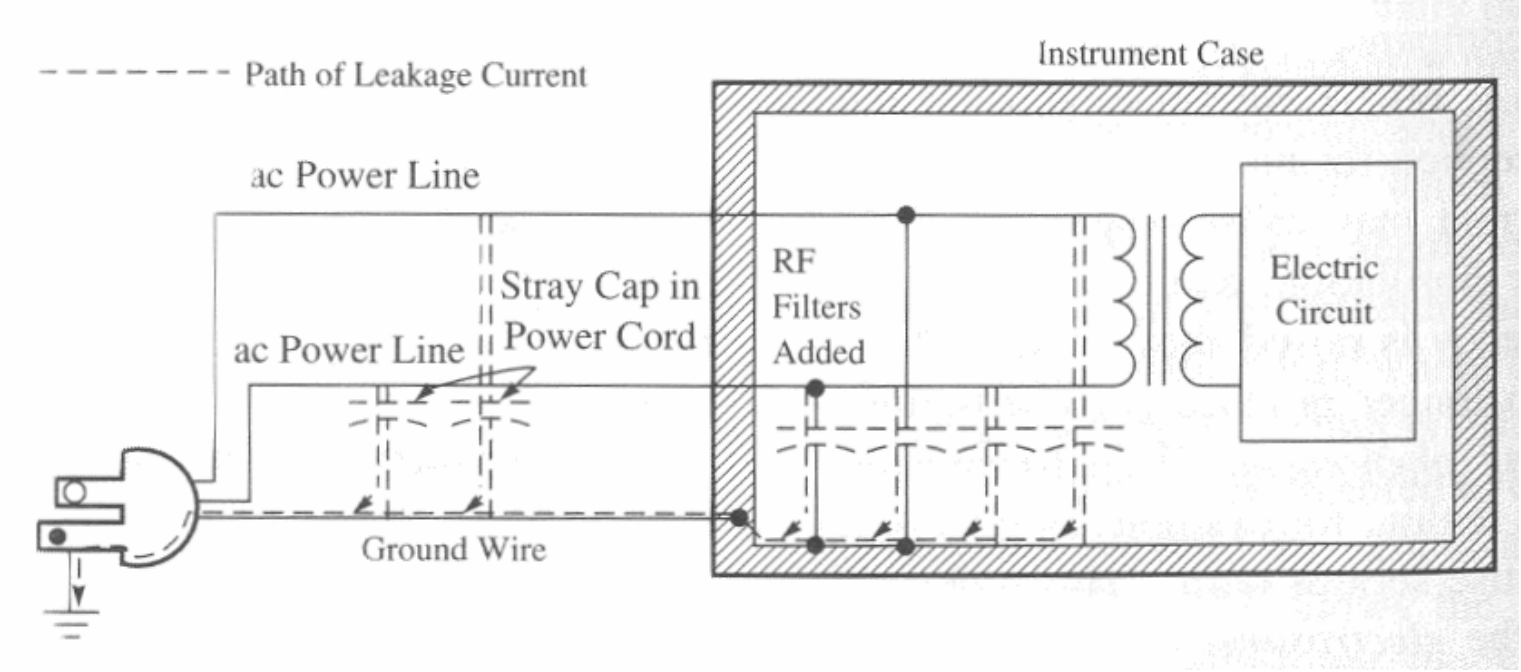


그림 15-6. 누설 전류를 증가시키는 RF 필터

누설전류 (Leakage current)

누설전류 100uA가 전력선으로부터 측정기 섀시로 흐르고 있다고 가정할 때

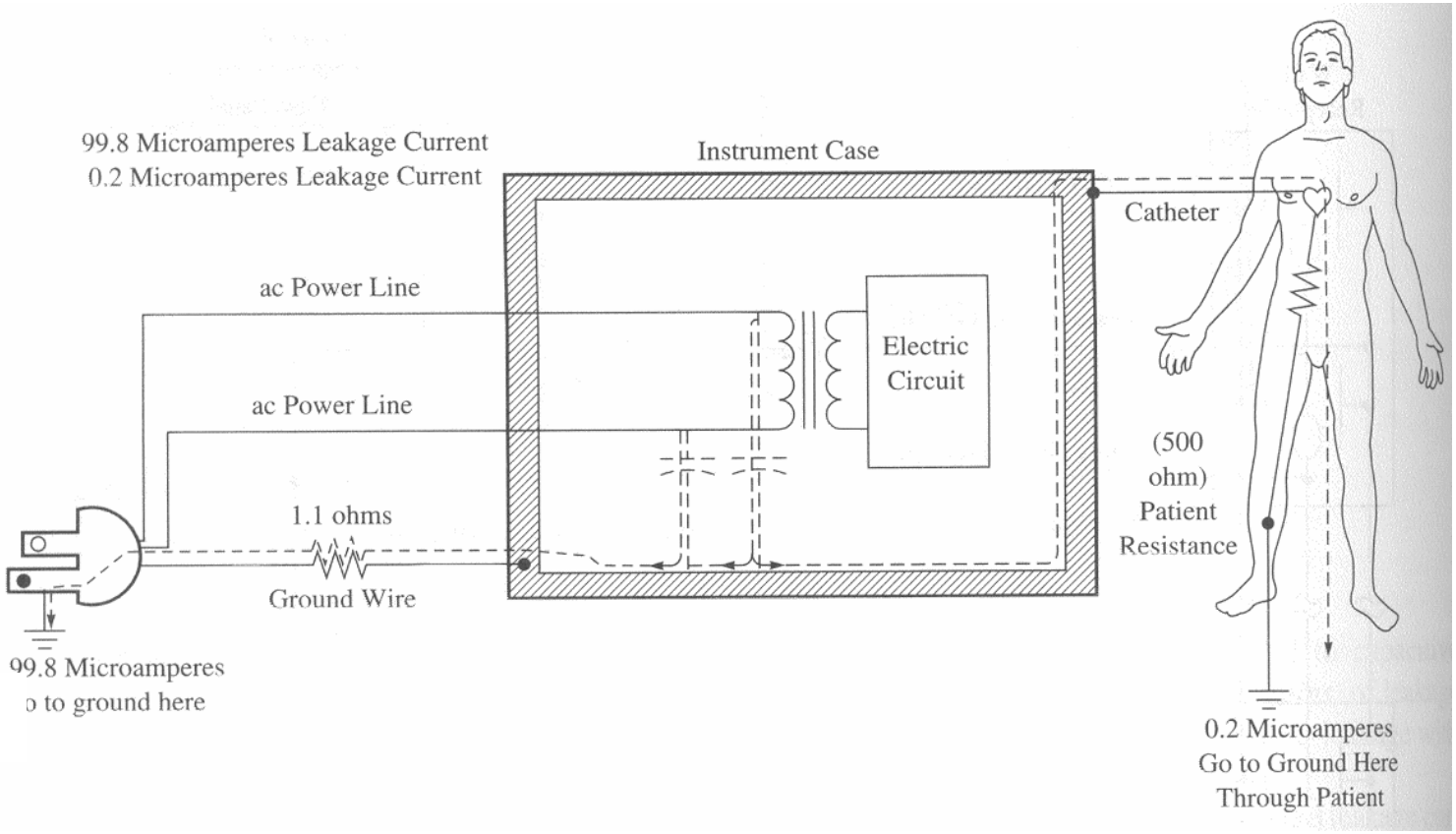


그림 15-8. 누설 전류에 의한 정상 경로

누설전류 (Leakage current)

누설전류 100uA가 전력선으로부터 측정기 섀시로 흐르고 있다고 가정할 때

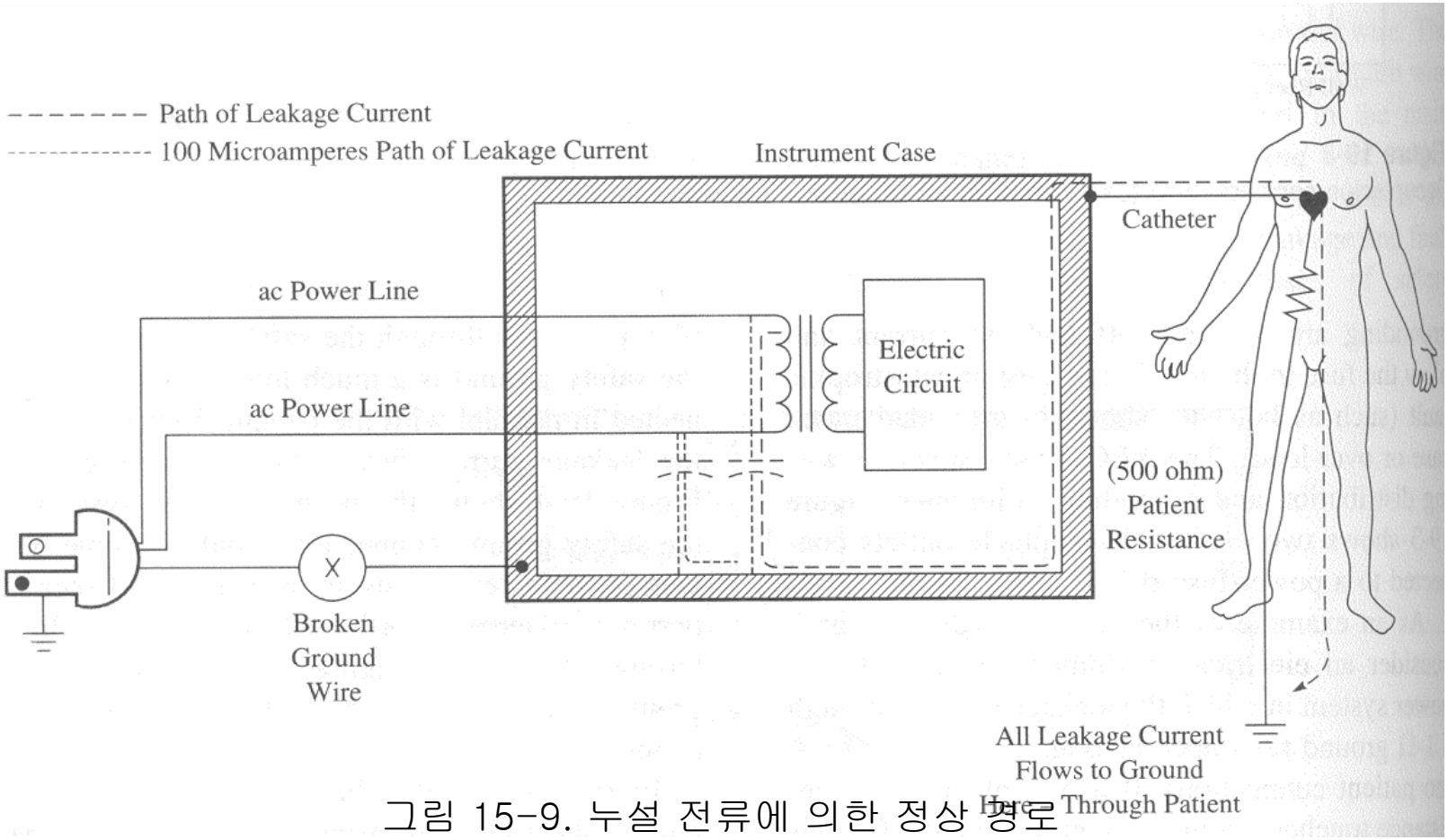


그림 15-9. 누설 전류에 의한 정상 경로

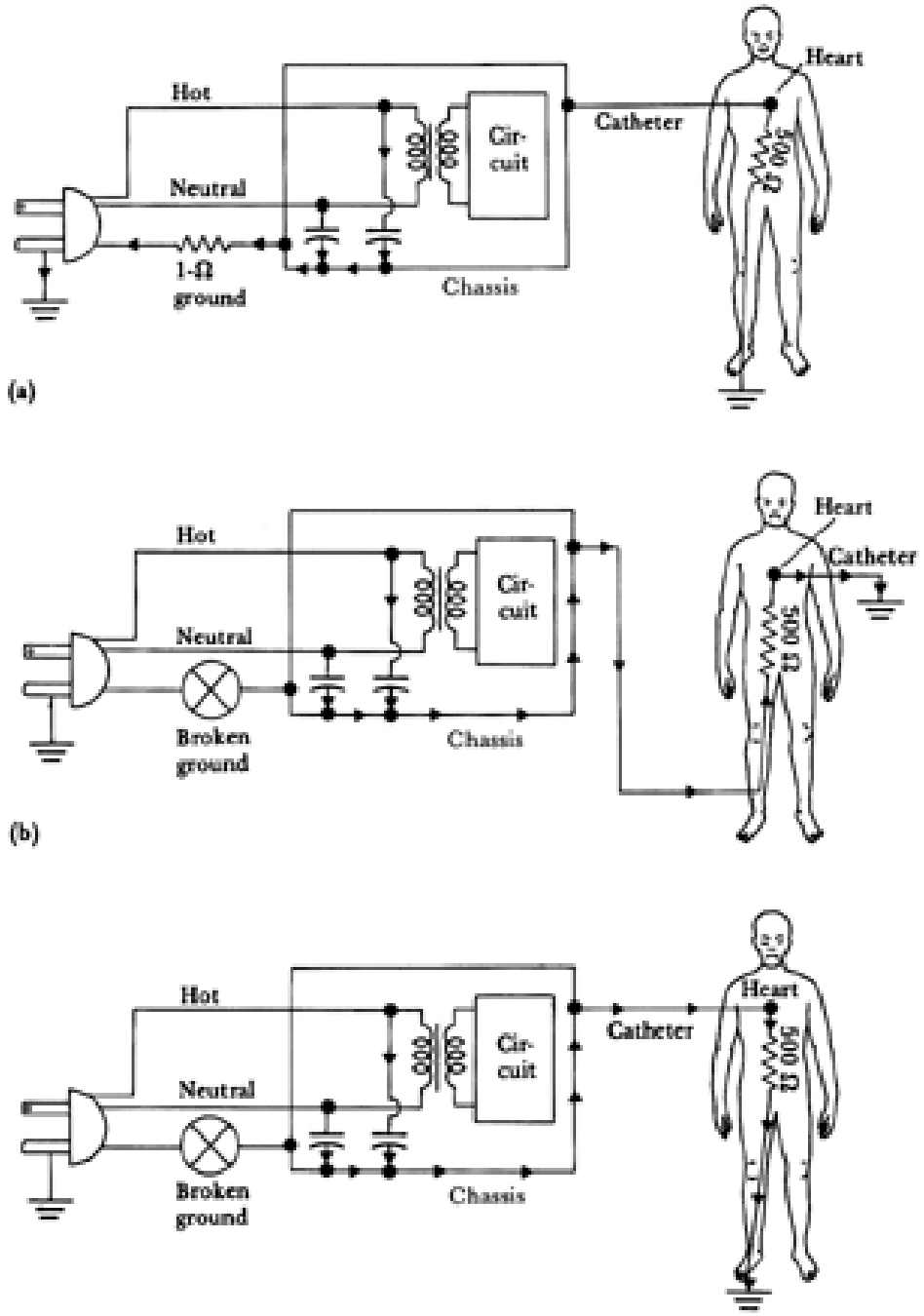
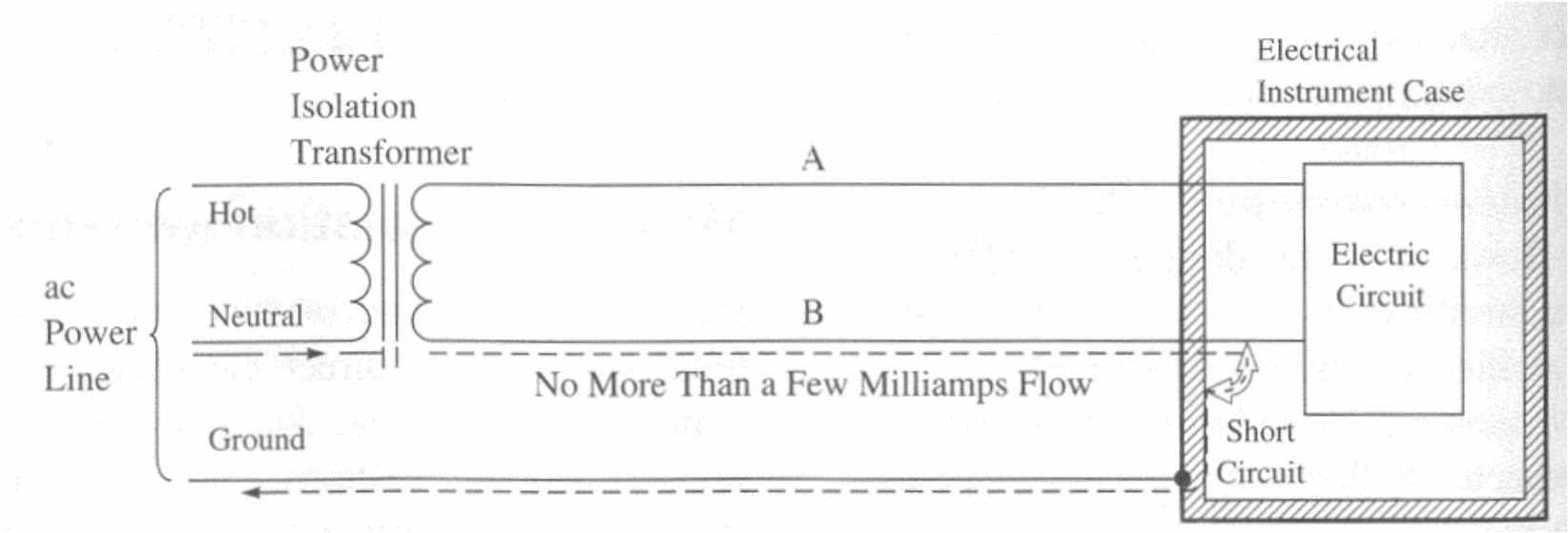


Figure. Assume 100 μA of leakage current from the power line to the instrument chassis. (a) Intact ground, and 99.8 μA flows through the ground. (b) Broken ground, and 100 μA flows through the heart. (c) Broken ground, and 100 μA flows through the heart in the opposite direction.

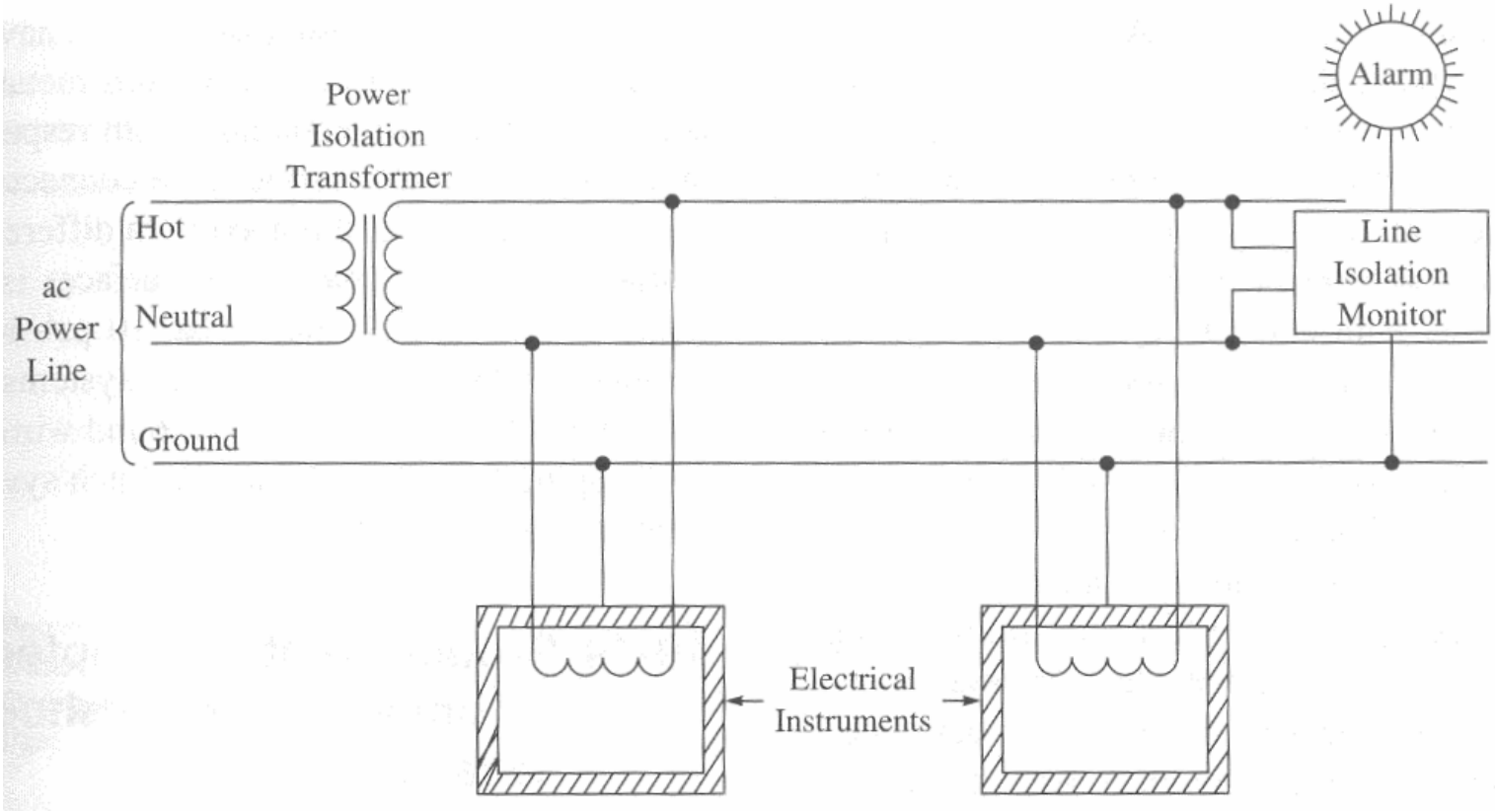
누설전류 (Leakage current)

- 고전적인 대처 방안
  - 3선 power line(safety ground wire)를 사용 (Figure 15-5, Figure 15-6 참조)
- 3선 라인 정의
  -
  -
  -

- LIM : Line isolation monitor
  – isolated power line과 접지간의 임피던스를 연속적으로 모니터링하는 장비



전력선 분리 시스템(LIM)



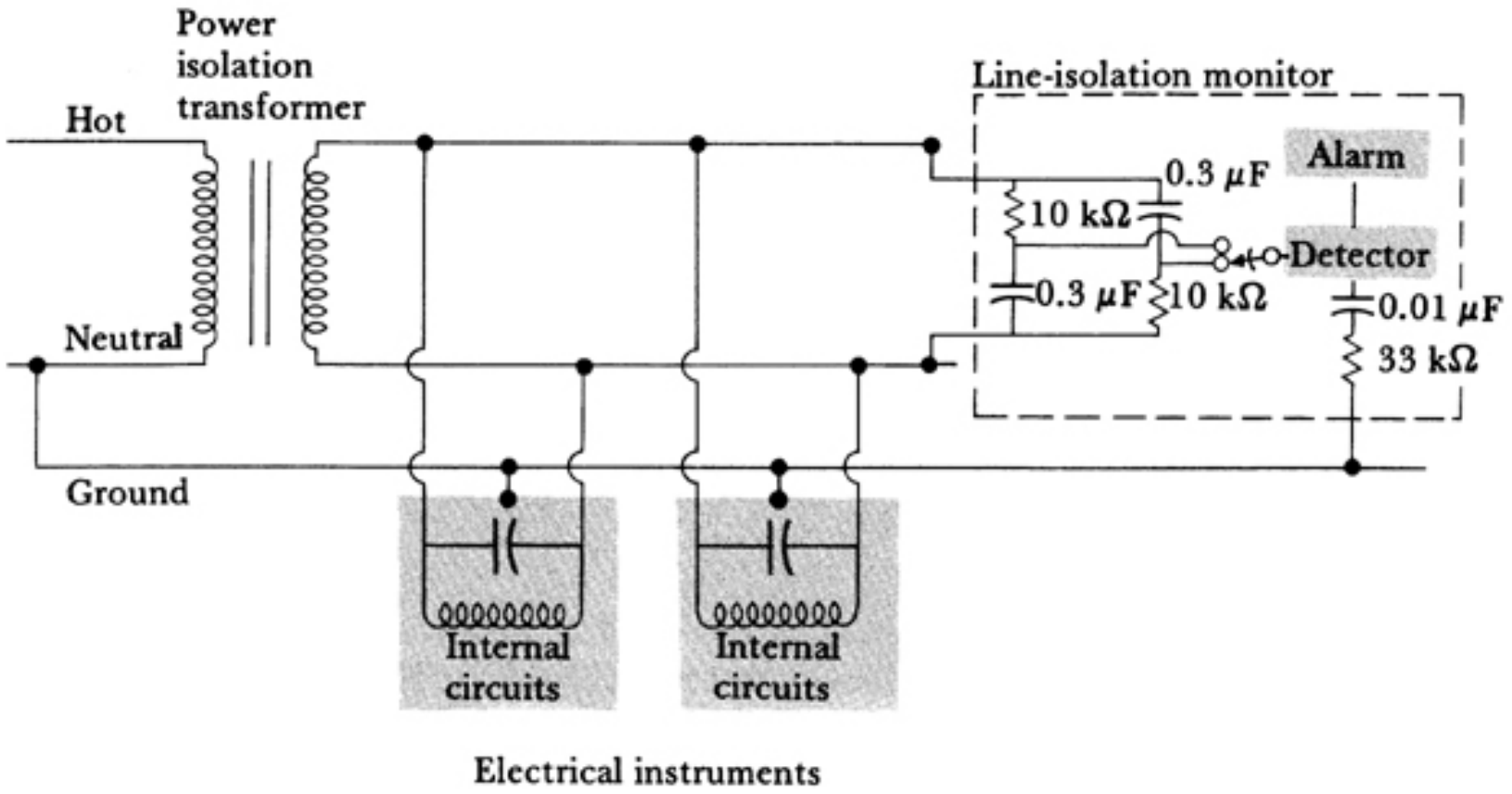
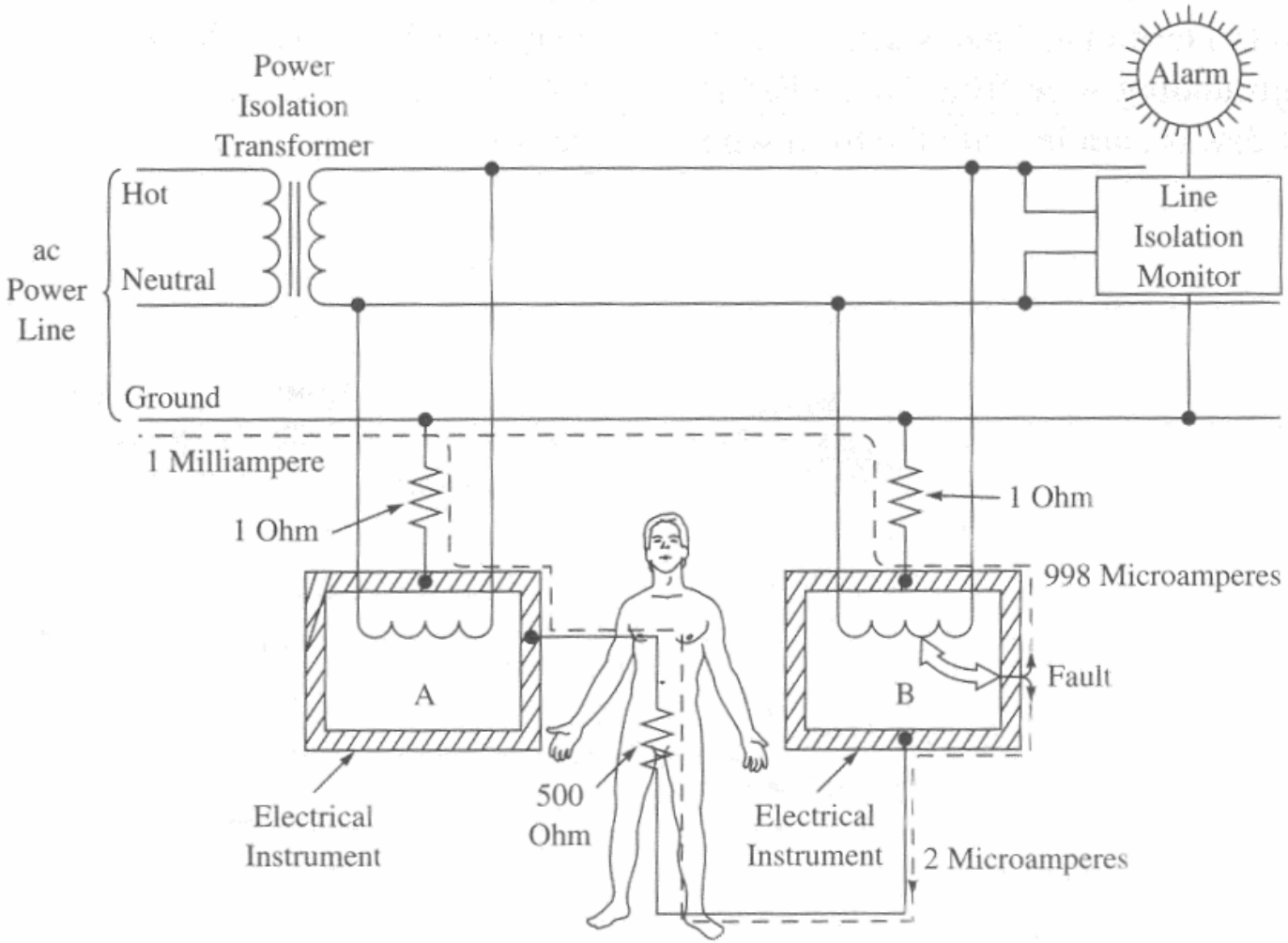
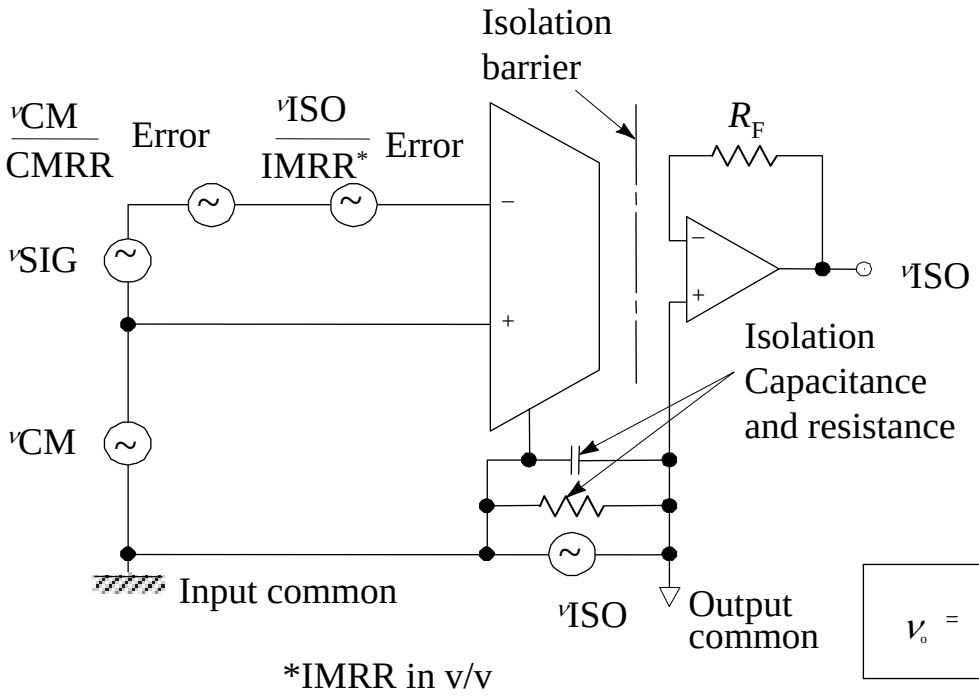


Figure. Power-isolation-transformer system with a line-isolation monitor to detect ground faults.

전력선 분리 시스템(LIM)

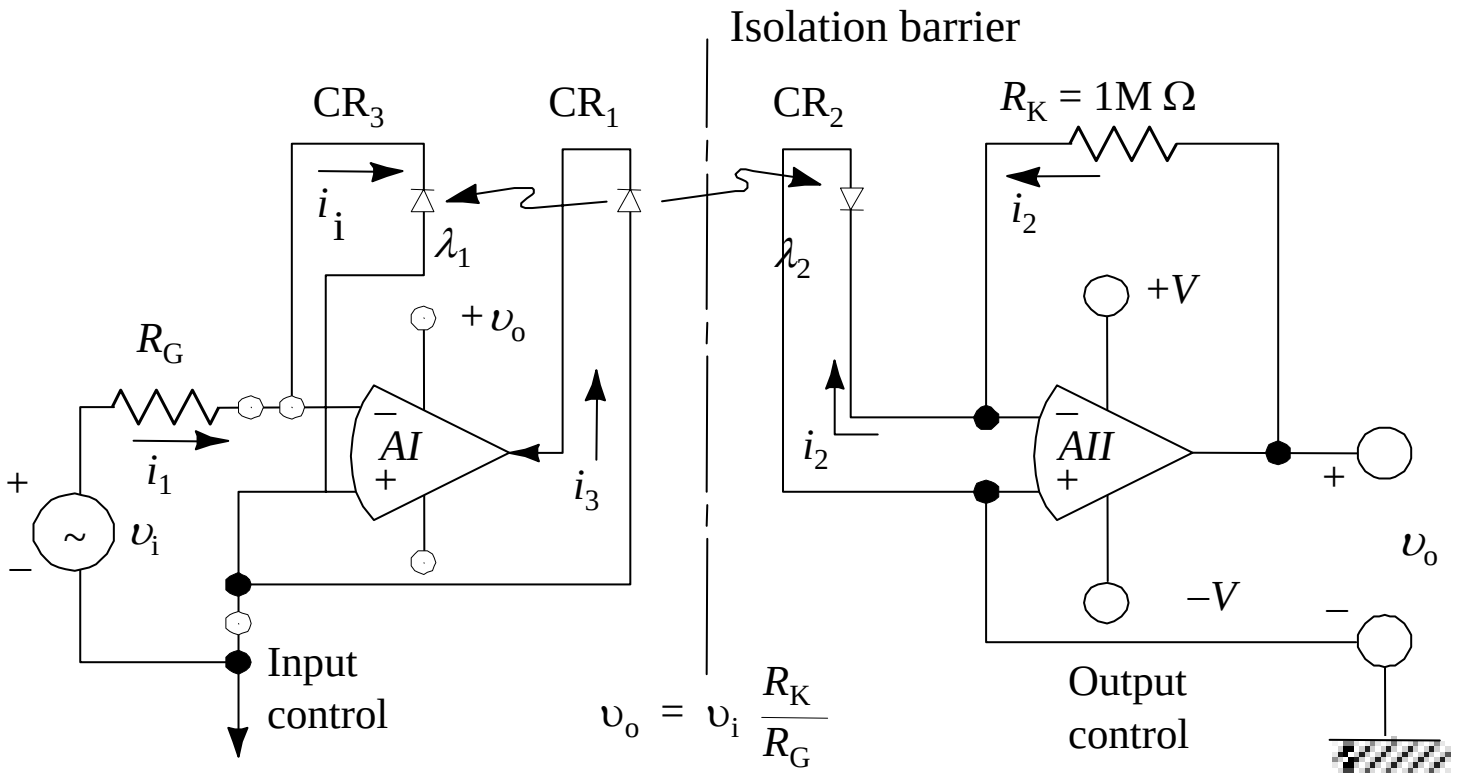


신호 분리 종류

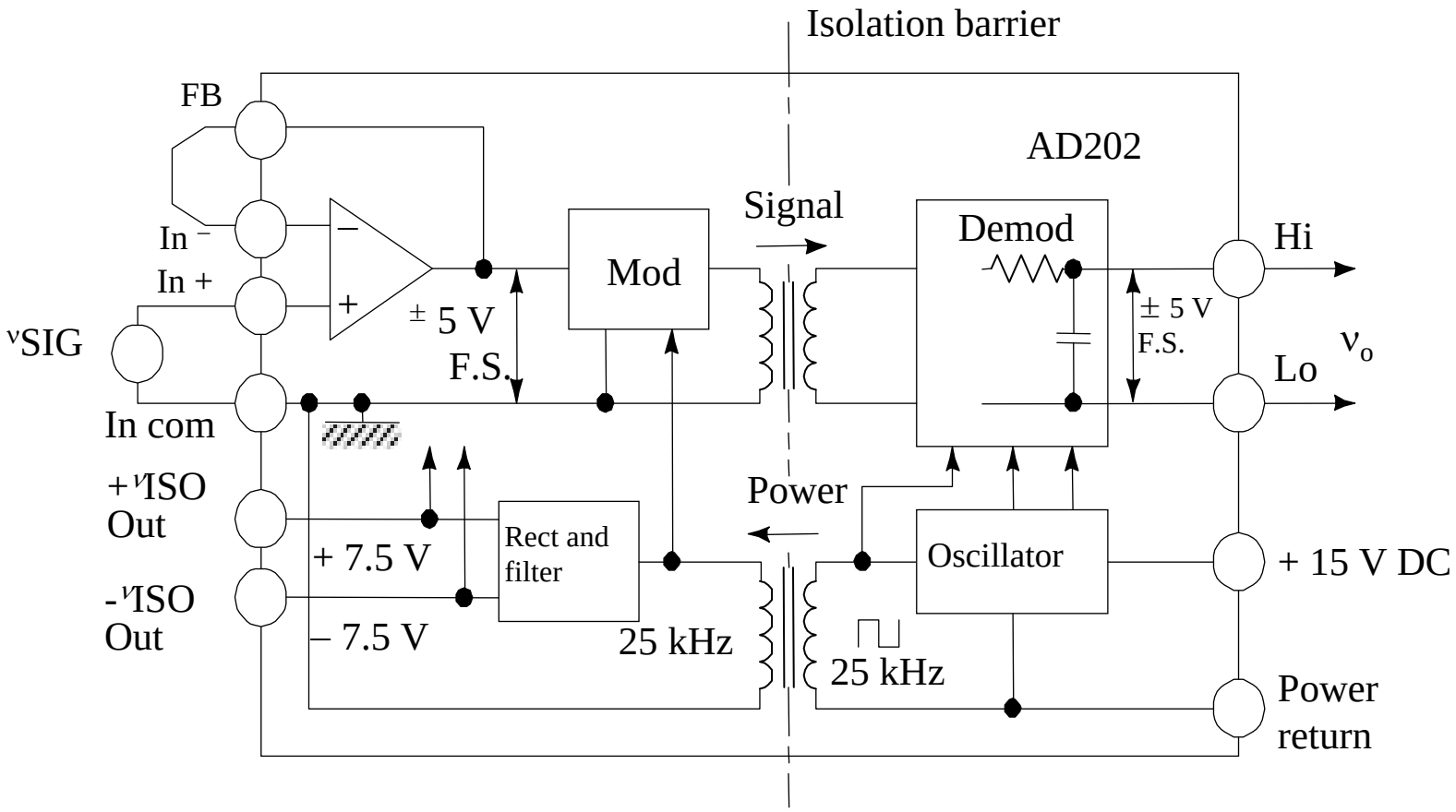


$$v_o = \left(v_{SIG} \pm \frac{v_{CM}}{CMRR} \pm \frac{v_{ISO}}{IMRR} \right) \text{Gain}$$

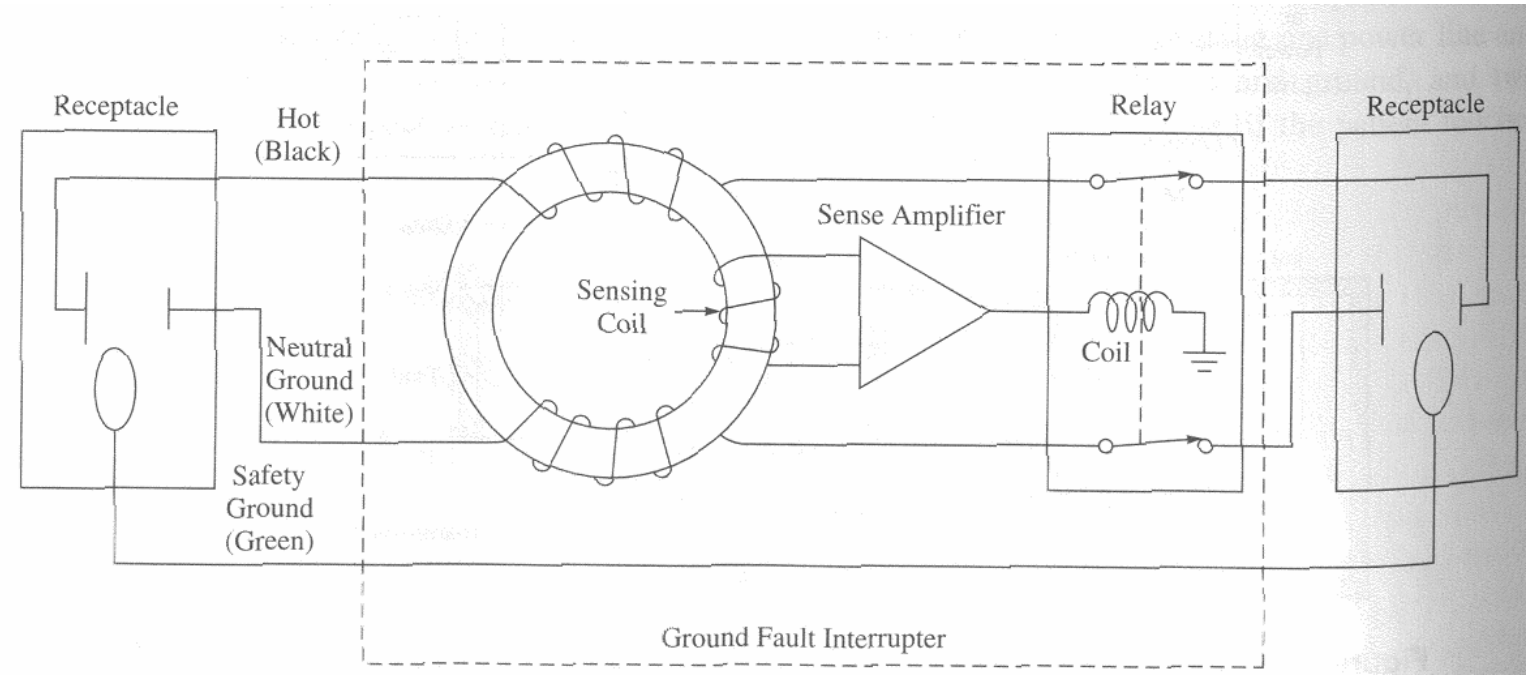
신호 분리 종류


Isolation barrier
CR3
CR1
CR2
RK = 1M Ω
i i
λ1
λ2
i2
+V
+ υo
RG
AI
AII
i1
i2
i3
+
υi
–
Input
control
υo = υi RK/RG
–V
Output
control
υo

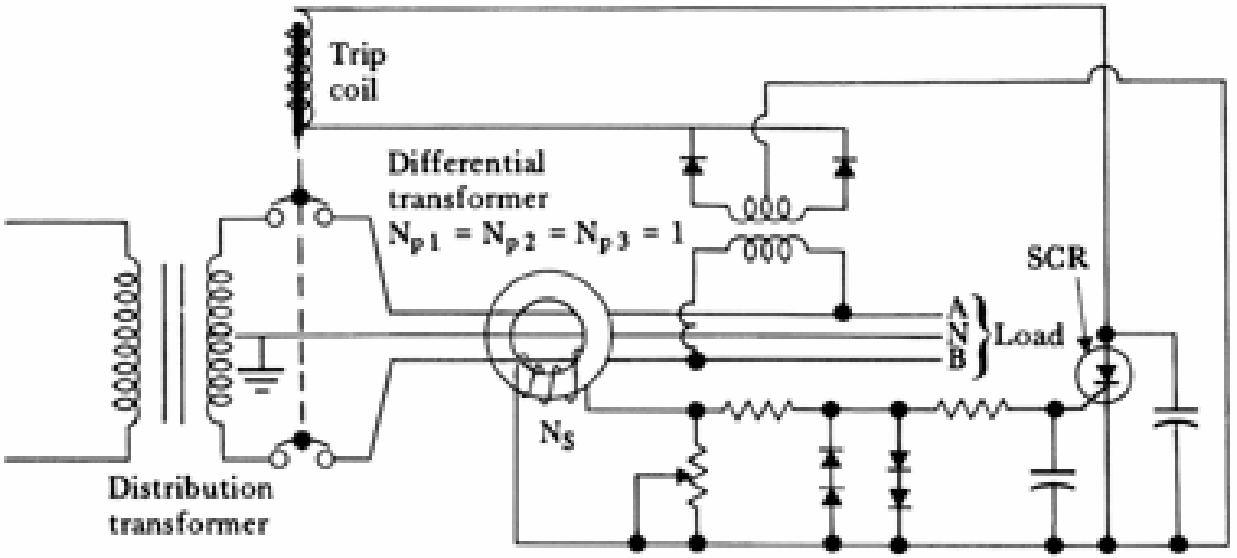
신호 분리 종류

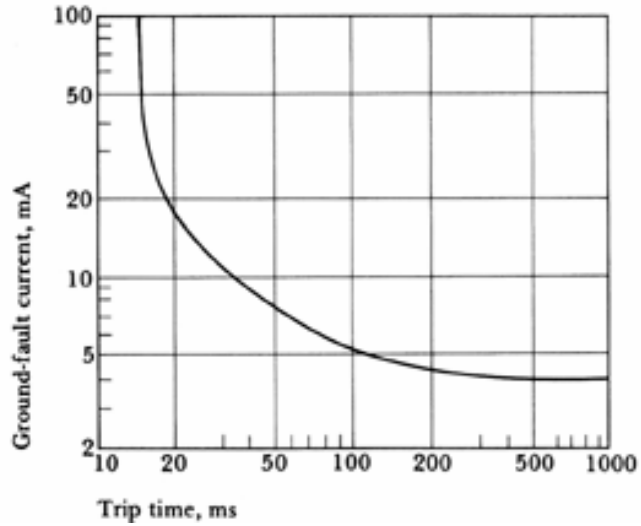

Isolation barrier
FB
AD202
Signal
Demod
Hi
In −
In +
Mod
± 5 V
F.S.
± 5 V
F.S.
VSIG
Lo
v_o
In com
+VISO
Out
Power
Rect and
filter
Oscillator
+ 15 V DC
+ 7.5 V
-VISO
Out
– 7.5 V
25 kHz
25 kHz
Power
return

- 전기 쇼크의 위험을 줄이기 위한 ground fault interrupters(GFI)
  -



Ground fault interrupters (GFI)





- 전기 안전 시험 장비


Advanced Test Equipment
©2000


PHENIX
ADVANCED TEST EQUIPMENT CORP © 2003

요약

- 전기 안정성
- 누설 전류
- 전력 분리 시스템
- 전기 쇼크의 위험을 줄이는 장비 접지 시스템
- 전기 쇼크의 위험을 줄이기 위한 적절한 전원선, 배치, 그리고 접지